SIS3204Y(5) Natural Language Processing Week 2: Text Processing | Revision Question Paper with Model Answers

University of Mauritius, Faculty of Information, Communication and Digital Technologies
Total: 100 marks | Suggested time: 2 hours

Instructions: Answer ALL questions. Marks are shown in brackets. The model answer (green) appears directly under each question; cover it while practising. Marks in brackets within answers show the marking scheme.

SECTION A: Definitions and Foundations (20 marks)

Q1. Define **text processing** and explain why it is needed before a computer can analyse human language. **[3]**

MODEL ANSWER

- **Text processing** is the set of steps that **cleans, transforms and represents** raw text in a structured form machines can analyse. (1)
- Human language is **flexible, context-dependent, noisy and ambiguous**. (1)
- Computers need **structured, numerical representations**; they work with numbers and patterns, not raw text. Text processing is the “bridge”. (1)

Q2. Define **tokenisation** and state the THREE kinds of units a token can be. **[3]**

MODEL ANSWER

- **Tokenisation** is the process of **breaking text into smaller units called tokens**. (1)
- Tokens can be **words, subwords or characters** (1 mark for two, 2 marks for all three). (2)

Q3. Define **stop words** and give FOUR examples. **[3]**

MODEL ANSWER

- **Stop words** are **common words that appear frequently but carry little meaning by themselves**; they are often removed to reduce noise. (1)
- Examples (any four, ½ mark each): **the, is, at, which, and, of, in, a**. (2)

Q4. Define **stemming** and give ONE example showing the stem produced. **[3]**

MODEL ANSWER

- **Stemming** reduces words to their **base or root form** by removing prefixes and suffixes using **heuristic (rule-based) methods**. (2)
- Example: **studying, studies, studied → studi**; or running → run, happily → happi. (1)

Q5. Define **lemmatisation** and give ONE example. **[3]**

MODEL ANSWER

- **Lemmatisation** reduces words to their **dictionary form (lemma)**, considering **part of speech and meaning**, using vocabulary and morphological analysis. (2)
- Example: **studying, studies, studied → study**; or mice → mouse, went → go. (1)

Q6. What is **text cleaning** (text normalisation)? State its goal. **[3]**

MODEL ANSWER

- **Text cleaning** is the first step in preprocessing; it **removes or standardises unwanted, noisy or inconsistent elements** from raw text. (2)
- **Goal:** convert messy text into a **clean and consistent form**. (1)

Q7. Define **lowercasing** and state ONE NLP task for which it may be unsuitable. **[2]**

MODEL ANSWER

- **Lowercasing** converts all uppercase letters to lowercase so that different cases of a word are treated as the same token. (1)
- Unsuitable where case carries meaning, e.g. **Named Entity Recognition (NER)**. (1)

SECTION B: Enumeration and Listing (20 marks)

Q8. State FIVE reasons why raw text is “messy”, giving a short example of each. **[5]**

MODEL ANSWER

- (1 mark each for any five; reason and example both needed)
- **Inconsistent capitalisation:** “NLP is fun”, “nlp is fun”, “NLP IS FUN”.
- **Punctuation and symbols:** “Wow!!! NLP is great!!!”, emojis.
- **Numbers in many formats:** “10/10”, “ten”, “10 out of 10”.
- **URLs, emails and mentions:** https://uom.ac.mu, info@uom.ac.mu, @john_doe.
- **Spelling errors and slang:** “I luv NLP”, “awsum”, “gr8”, “thnx”.
- **Extra spaces and line breaks:** irregular spacing in “NLP is powerful”.

Q9. List FOUR common tokenisation methods and describe each in one line. **[4]**

MODEL ANSWER

- (1 mark each)
- **Whitespace tokenisation:** split by spaces.
- **Word tokenisation:** split into words and remove punctuation.
- **Subword tokenisation (BPE, WordPiece):** split words into subwords to handle rare or unknown words.
- **Character tokenisation:** split into individual characters.

Q10. State THREE benefits of removing stop words. **[3]**

MODEL ANSWER

- (1 mark each for any three)
- Reduces **vocabulary size and dimensionality**.
- Improves **efficiency and speed** and reduces storage.
- Improves **information retrieval and search**; helps focus on **meaningful content words**.
- Improves **model performance** in many tasks.

Q11. List FIVE common text cleaning tasks. **[5]**

MODEL ANSWER

- (1 mark each for any five)
- Convert to **lowercase**.
- Remove **punctuation and special characters**.
- Remove or standardise **numbers**.
- Remove **URLs, emails and HTML tags**.
- Remove **hashtags, mentions and emojis**.
- Remove **extra spaces and line breaks**.

Q12. Name THREE real-world applications of NLP and give ONE example tool or product for each. **[3]**

MODEL ANSWER

- (1 mark each: application + example)
- **Chatbots and virtual assistants:** ChatGPT, Siri, Google Assistant.
- **Search engines:** Google Search, Bing.
- **Machine translation:** Google Translate, DeepL, Microsoft Translator.

- Also accepted: text summarisation, sentiment analysis, spam detection, document understanding, speech recognition.

SECTION C: Explanatory and Conceptual Questions (24 marks)

Q13. Explain why lowercasing improves NLP processing, using the words “NLP”, “Nlp” and “nlp” as an example. State one case where lowercasing can cause problems. **[4]**

MODEL ANSWER

- Without lowercasing, **NLP, Nlp and nlp are treated as different words**; with lowercasing they all map to **nlp**. (1)
- Benefits: **reduces vocabulary size**, **improves matching and retrieval**, **helps models generalise better**. (2)
- Problem case: **NER**, where capitalisation signals names (proper nouns). (1)

Q14. Explain the THREE techniques used to handle spelling, contractions and informal language. Give an input and output example for each, and explain why contractions should be expanded **before** tokenisation. **[6]**

MODEL ANSWER

- **Spelling normalisation:** fix common mistakes. *recievd the application → received the application.* Tools: PySpellChecker, SymSpell, domain dictionary. (1)
- **Expanding contractions:** *I’m happy, she’s not here → I am happy, she is not here* (don’t → do not, can’t → cannot, won’t → will not). (1)
- **Handling informal language:** convert slang to standard form with a **mapping list or glossary**. *u r goin 2 the mtg tmr → you are going to the meeting tomorrow.* (1)
- **Why expand before tokenisation:** otherwise tokenisers may split **“don’t” into “don” and “t”**, creating meaningless tokens. (2)
- Examples for each technique carry the remaining mark. (1)

Q15. Explain how **subword tokenisation** works, using the word “internationalization” as an example. Explain why it handles rare or unknown words better than word-level tokenisation. **[6]**

MODEL ANSWER

- Subword tokenisation breaks words into **smaller meaningful units** (e.g. WordPiece, BPE) rather than whole words. (1)
- Example: *internationalization → inter, ##national, ##ization*; a rare word is split into known pieces using the algorithm’s rules. (2)
- Special tokens such as **[CLS] and [SEP]** may be added for model input. (1)
- Handles **rare and unknown words** by composing them from known pieces, whereas word-level tokenisation **fails if the word is unknown (OOV)**. (1)
- Other advantages: **smaller vocabulary**, **better generalisation** (prefixes, suffixes, roots), **shorter sequences** than character-level models. (1)

Q16. Using TWO examples, explain how removing stop words can change the meaning of a sentence. **[4]**

MODEL ANSWER

- (2 marks per example: original, result, effect)
- *“Where is the nearest bank?”* → **“nearest bank”**: the question word **“where”** is lost, so it is no longer a question about location.
- *“She is not happy.”* → **“happy”**: **negation is removed** and the meaning becomes the opposite.
- Also accepted: “I agree with you” → “agree”; “I saw him because he was late” → “late”.

Q17. Explain what **sentence tokenisation** is and describe FOUR challenges that make it difficult. **[4]**

MODEL ANSWER

- **Sentence tokenisation** splits a paragraph or document into individual sentences, helping systems understand **structure and boundaries**. (1)
- Challenges (1 mark each for any three): **abbreviations** (Dr., Mr., U.S.A.) causing wrong splits; **exclamations and questions**; **ellipses (...) and multiple punctuation**; **newlines vs real sentence breaks**; **domain-specific writing styles**.

SECTION D: Comparative Questions (12 marks)

Q18. Compare **stemming** and **lemmatisation** in a table using FOUR criteria. Give examples for both. **[6]**

MODEL ANSWER

Criterion	Stemming	Lemmatisation
Method	Simple rule-based heuristics (prefix/suffix removal)	Vocabulary and morphological analysis with POS and context
Output	May produce **non-words** (e.g. studi)	Produces **valid dictionary words** (e.g. study)
Speed	**Faster**, mostly language independent	**Slightly slower**, language dependent
Accuracy	Less accurate, may lose meaning	More accurate, preserves meaning and context

- ½ mark per cell, 4 criteria = 4 marks. (4)
- Stemming examples: running → run, studies → studi, relational → relat, happily → happi. (1)
- Lemmatisation examples: studies → study, relational → relation, mice → mouse, children → child, went → go. (1)
- Use lemmatisation for search, QA and summarisation; stemming for quick tasks where exact meaning is less critical.

Q19. Compare **word-level**, **subword (BPE)** and **character-level** tokenisation in the context of modern LLMs, using the text “Internationalization makes NLP powerful!”. **[6]**

MODEL ANSWER

Approach	Output for the sample text	Assessment
Word-level	Internationalization, makes, NLP, powerful, !	Simple, **not used in LLMs**; **fails for unknown or rare words**
Subword (BPE)	Intern, ational, ization, make, s, NLP, power, ful, !	**Used in most LLMs**; handles any word and captures meaningful pieces
Character-level	I, n, t, e, r, n, a, t, i, o, n ...	**Too many tokens**, slower and less efficient

- Table: 1 mark per row plus 1 for headings/clarity. (3 to 4)
- Subword tokenisation balances vocabulary size, coverage and efficiency; used in GPT, LLaMA, Gemini and Mistral. (1)
- Subword works across **multiple languages and scripts** and supports code, names, slang and typos. (1)
- Fewer tokens per sentence reduce sequence length and speed up training and inference. (1)

SECTION E: Process and Pipeline Questions (14 marks)

Q20. Describe the six stages of the **text preprocessing pipeline**. Trace the sample text *“Wow!!! NLP is awesome 😁. Visit https://uom.ac.mu now!!!”* through the stages. **[8]**

MODEL ANSWER

- Marking: up to 4 for stage descriptions, up to 4 for the traced example.
- **1. Cleaning:** remove noise (lowercasing, punctuation, numbers, URLs, HTML).
- **2. Tokenisation:** split text into tokens (words, subwords or sentences).

- **3. Stop word removal:** remove common words with little meaning (the, is, at, which).
- **4. Stemming / Lemmatisation:** reduce words to base or root form to group similar words.
- **5. Feature extraction:** convert text into numerical features (BoW, TF-IDF, embeddings).
- **6. Ready for NLP models:** classification, clustering, etc. The output of one step is the **input of the next**.

Traced example

Stage	Output
Raw text	Wow!!! NLP is awesome 😀. Visit https://uom.ac.mu now!!!
After cleaning	**wow nlp is awesome visit uom ac mu now**
After tokenisation	[wow, nlp, is, awesome, visit, uom, ac, mu, now]
After stop words	[wow, nlp, awesome, visit, uom, ac, mu, now] ("is" removed)
After stemming / lemmatisation	[wow, nlp, awesom, visit, uom, ac, mu, now]
After feature extraction	A numerical vector, e.g. [0.12, 0.00, 0.31, 0.05, ..., 0.00]

Q21. Describe FOUR metrics used to evaluate text preprocessing. For each, state what it measures, how it is calculated and what a good result looks like. **[6]**

MODEL ANSWER (any four, 1.5 marks each)

Metric	Measures	Calculation	Good result
Token Count Reduction	Reduction in tokens after preprocessing	(1 − New/Original) × 100	Moderate to high (20% to 60%)
Unique Token Ratio (UTR)	Proportion of unique tokens	Unique Tokens / Total Tokens	Higher is better (> 0.4)
Information Retention	Important content kept	Compare key entities, keywords or labels (manual or automatic)	High retention (> 90%)
Stop Word Removal %	Proportion of tokens removed as stop words	Stop Words Removed / Original Tokens × 100	Depends on domain (10% to 40%)
Processing Time	Time taken to preprocess	Seconds for a dataset	Lower is better

- Lecture example: total tokens 15,200 → 9,100 (**−40.1%**).

SECTION F: Synthesis Essay (10 marks)

Q22. "There is no one-size-fits-all preprocessing." Critically discuss this statement with reference to stop word removal, lowercasing, punctuation removal, and the choice between stemming and lemmatisation. Include best practices for designing a preprocessing pipeline. **[10]**

MODEL ANSWER

- **Thesis (2):** preprocessing steps help only if they match the **task, data and model**. Good preprocessing is "the right steps for the right data and the right task": **clean, but not too clean**.
- **Stop words (2):** remove for content and topic tasks (**document classification, clustering, topic modelling**, BoW/TF-IDF). Keep for **sentiment analysis** ("not", "no"), **NLI**, **question answering** (where, when, how) and transformer models relying on word order. Example: "Where is the nearest bank?" → "nearest bank" loses meaning. Information retrieval: **sometimes**.
- **Lowercasing and punctuation (2):** lowercasing reduces vocabulary but harms **NER**. Punctuation removal reduces noise, but **exclamation marks can carry emotion in sentiment analysis**. Numbers may matter (e.g. product reviews). Do not remove information important to the task.
- **Stemming vs lemmatisation (2):** stemming is fast but may give non-words (studi) and suits quick tasks. Lemmatisation is more accurate (better → good with the right POS tag) and is preferred for search, QA and high-precision tasks.
- **Best practices (2):** understand task and data first; start simple; **inspect intermediate outputs**; apply the **same pipeline to train, validation and test data**; **evaluate impact** (e.g. with and without stop words); keep a copy of raw data; use domain-specific stop word lists; make the pipeline configurable and **document it**. Avoid over-processing, inconsistent processing and ignoring edge

cases.

END OF REVISION PAPER